

SOLUTION REQUIREMENTS

Project Name: ComicCraft – AI Powered Comic Story Generator

Team ID: _____

Step 1: Functional Requirements (FR)

S.No	Requirement Category	Requirement Description	Priority
1	Authentication	Users can securely sign up, log in, and manage their accounts to save generated comics and preferences.	High
2	Authorization Levels	Regular users can generate and download comics. Administrators can monitor usage, manage users, and maintain system settings.	Medium
3	External Interfaces	The system integrates with Google Gemini APIs for story generation and AI image-generation services for comic illustrations.	High
4	Transaction Processing	The system processes user prompts, generates story content, creates comic images, and assembles them into downloadable comics.	High
5	Reporting	The system provides comic previews, generation history, download records, and user activity summaries.	Medium
6	Business Rules	Every comic must contain a structured storyline, multiple comic panels, AI-generated illustrations, and dialogues before export. The generated content must follow the selected tone and style.	High
7	Compliance to Laws or Regulations	The system must comply with data privacy regulations and ensure responsible AI content generation without harmful or inappropriate outputs.	Medium
8	Other Functional Requirements	Users can select comic genres, art styles, character descriptions, and export comics in PDF format.	Medium

Step 2: Non-Functional Requirements (NFR)

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
1	Performance & Speed	The system should generate comic stories and illustrations efficiently.	Story generation within 10 seconds and complete comic generation within 60 seconds.
2	Scalability	The platform should support increasing numbers of users and	Supports at least 1,000 concurrent users.

S.No	NFR Category	Requirement Description	Target Metric / Acceptance Criteria
		comic-generation requests.	
3	Security & Data Privacy	User information and generated content must be protected using secure communication and storage mechanisms.	HTTPS enabled, encrypted data storage, secure authentication.
4	Reliability & Availability	The platform should remain available and stable during normal operation.	Minimum 99.5% uptime per month.
5	Usability & Accessibility	The system should be easy to use for beginners and accessible across devices.	Responsive interface with intuitive navigation and accessibility support.
6	Maintainability	The codebase should be modular, documented, and easy to update.	Modular architecture with clear documentation and reusable components.
7	Portability	The application should run across different operating systems and deployment environments.	Compatible with Windows, Linux, macOS, and cloud deployment platforms.
8	Compatibility	The system should support modern web browsers and mobile devices.	Works on Chrome, Edge, Firefox, Safari, Android, and iOS browsers.

Functional Requirements Summary

ComicCraft must allow users to enter story prompts, generate AI-powered comic storylines, create illustrations for each panel, combine text and images into a comic layout, preview the generated comic, and download it as a PDF. The platform should integrate seamlessly with external AI services while ensuring secure user access and reliable performance.

Non-Functional Requirements Summary

ComicCraft must provide fast response times, high availability, strong security, scalability, and an intuitive user experience. The platform should be maintainable, portable, and compatible with multiple devices and operating systems to ensure long-term usability and growth.

Conclusion

The Solution Requirements define both the functional capabilities and quality standards expected from ComicCraft. These requirements serve as the foundation for system design, development, testing, and deployment, ensuring that the final solution effectively addresses the needs of storytellers, students, educators, and content creators.